

tion (green) is delayed by two hours with respect to the east (blue). The combined power is shown in red. By having two such plants, we can extend the generation of solar power for an additional two hours. To achieve this, a bidirectional transmission line between these two plants is necessary.

With such vast distance between these two locations, wind patterns will also differ. Hence, it is possible that when wind is strong in one location, the other location may have less wind, and vice versa. This difference is useful for achieving uniform power generation from windmills. Thus, the combined effect of solar and wind generation at two geographically different locations will help in generating steadier power for 24 hours.

Why HVDC

The distance between the east and west extreme points is 2933km. For such a long distance, only HVDC lines are suitable. HVDC connects two terminal stations. At these stations, AC is converted to DC, then transmitted over the line, received, and converted back to AC. The advantages of HVDC transmission include:

1. For very long distances (beyond breakeven distance), it transmits power very economically
2. Reduced power losses due to the absence of reactive power, absence of skin effect, and no dielectric losses
3. Thinner and cheaper conductors. Only two conductors are required, unlike three in HVAC lines, hence a narrower right of way, reducing land acquisition requirements
4. Full control over power flow
5. Easily integrates with smart grid

A major limitation of HVDC is that it transmits bulk power between two terminals only. And, to cover such a long distance, the initial investment required is much higher. To keep the initial investment on the lower side and to maximise utilisation, consider

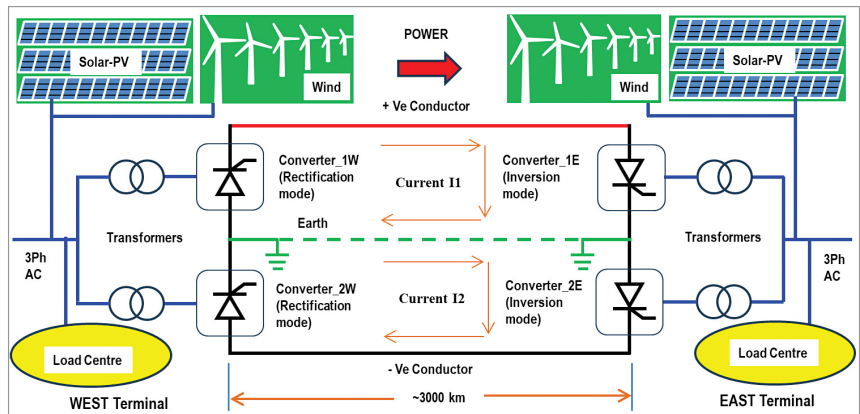


Fig. 2: Proposed power corridor with HVDC link connecting east and west terminals

these points:

1. The proposed HVDC line should be installed on a fixed latitude throughout the length, without any deviation (parallel to the equator). Such a line will be the shortest one and ensure the maximum time difference between the two stations.
2. The two terminals of this corridor should have the potential to set up solar and wind farms.
3. The generation cost of solar-PV and wind power is decreasing. Hence, this power should be utilised to avoid the use of 'peaker' power plants which use oil, and whose generation cost is much higher. This criterion should be followed while deciding the direction of power flow at any given time of the day.

Fig. 2 shows the block diagram of proposed power corridor. The bipolar HVDC link consists of Converter_1W and Converter_2W on the west terminal side. Similarly, on the east side Converter_1E and Converter_2E are installed. Converter_1W is connected to Converter_1E. Converter_2W is connected to Converter_2E. When power is to be transferred from west to east terminal, the west converters are operated in rectification mode to convert AC to DC. The east converters are operated in inversion mode to convert back DC to AC. I1 and I2 depict the currents flowing through each circuit. Note that, the current flowing through

earth is the difference between currents I1 and I2, which are in opposite directions. Usually, the system is operated in a balanced mode such that I1 is equal to I2, to ensure zero current flows through the earth terminals.

Whenever the power flow must be reversed, the operating modes of these converters are reversed by changing the firing angle of the power devices (thyristors or IGBTs).

All the converter inputs are connected to AC grid through 3-phase transformers to get the required DC voltage level. Higher the DC voltage, lower will be the line losses. Long distance bipolar links are usually designed to operate at around $\pm 600\text{kV}$ or more.

On the AC side of each terminal, power is fed from solar-PV and wind farms. Also, there will be a load centre (city or industrial area) which consumes power.

HVDC vs grid batteries

As shown in Fig. 1, the power output of each plant provides output $> 80\%$ for about 3.5 hours. Assuming that during this period, the power is being supplied to one load centre only, then the total duration for which $> 80\%$ output is available is 5.5 hours. This means the system works as a battery for two hours. Assuming the average power in this duration is 2GW, the storage capacity works out to be 4GWh, which is a large capacity. Table 2